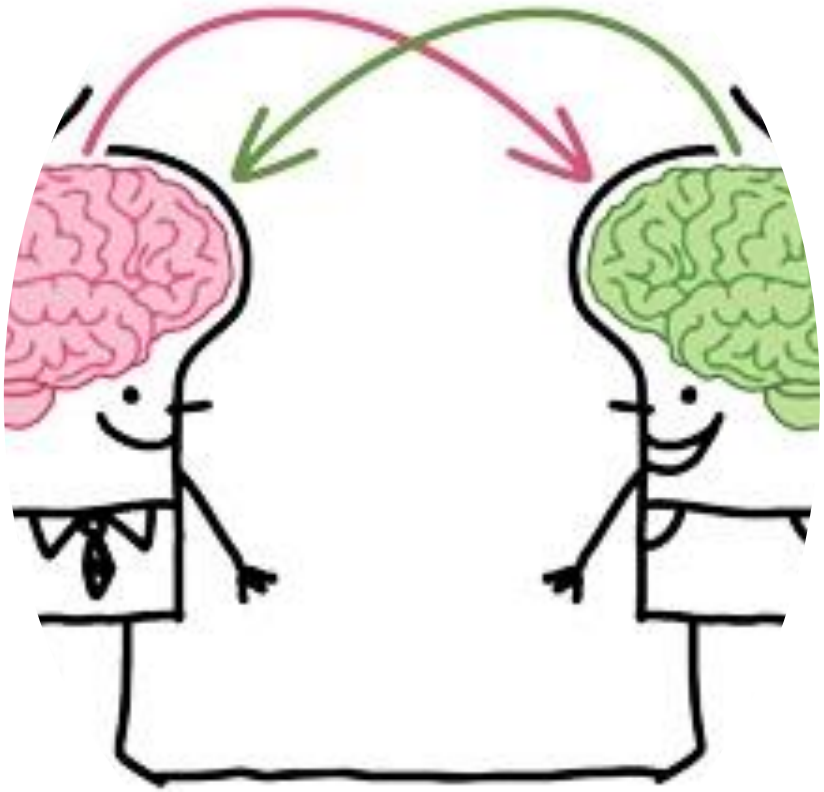
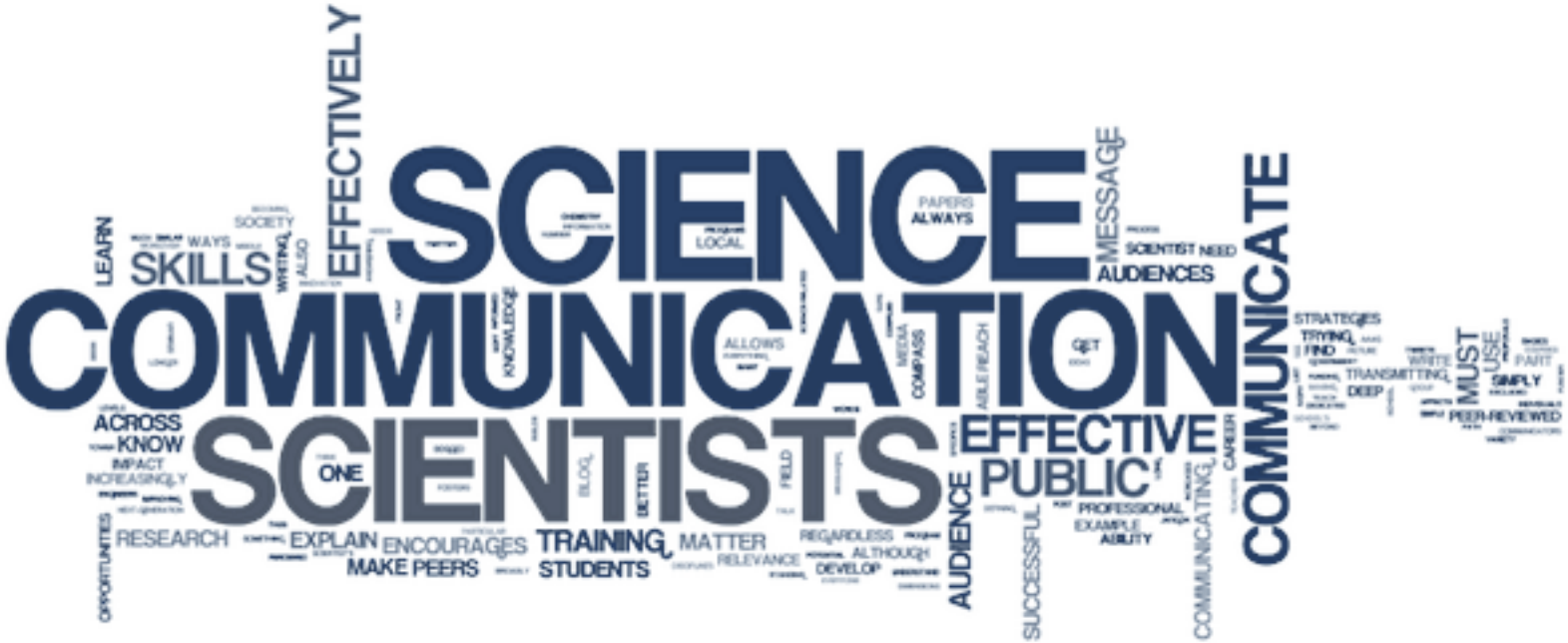
# Data Visualization and Plots in R

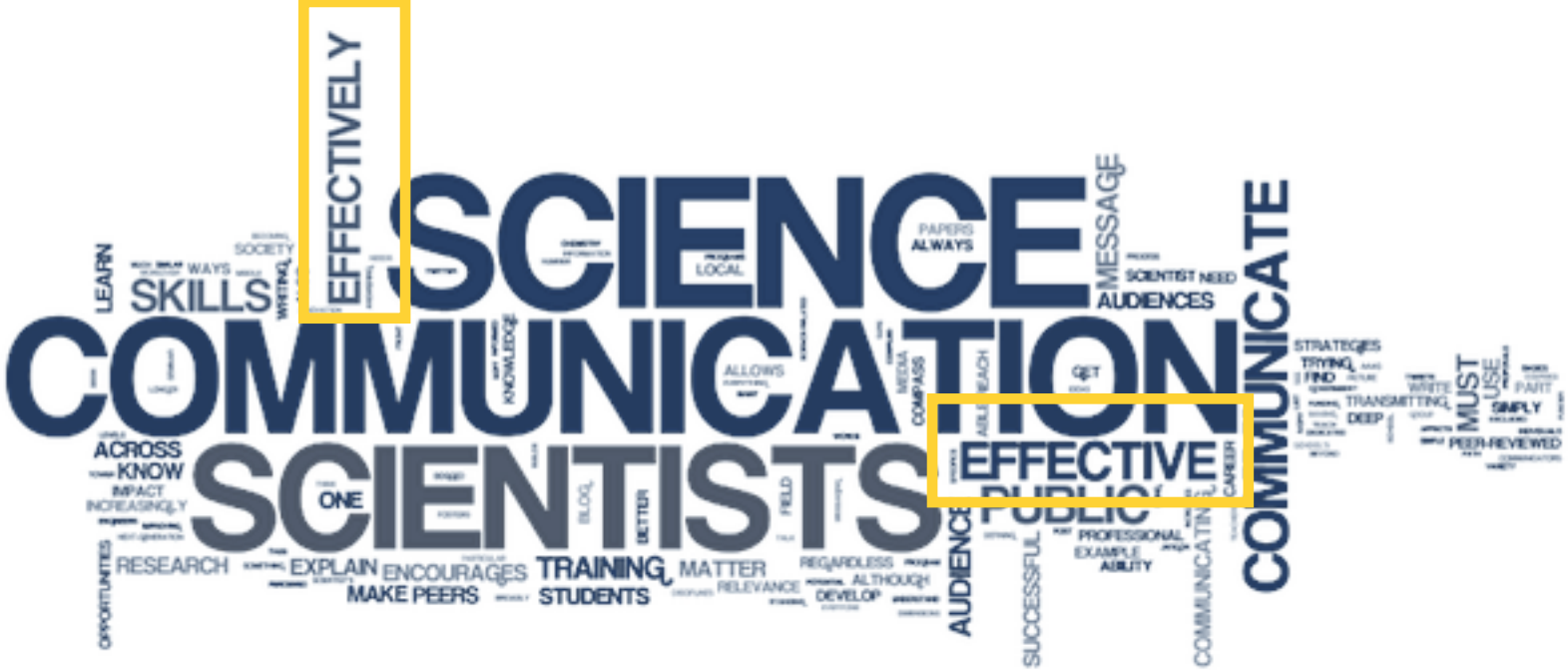
Carolina Sarmiento, USF
October, 2026



Objective:
**Communicate our science**




- Papers and proposals
- Talks
- Teach
- Outreach
- Social media + Journalists



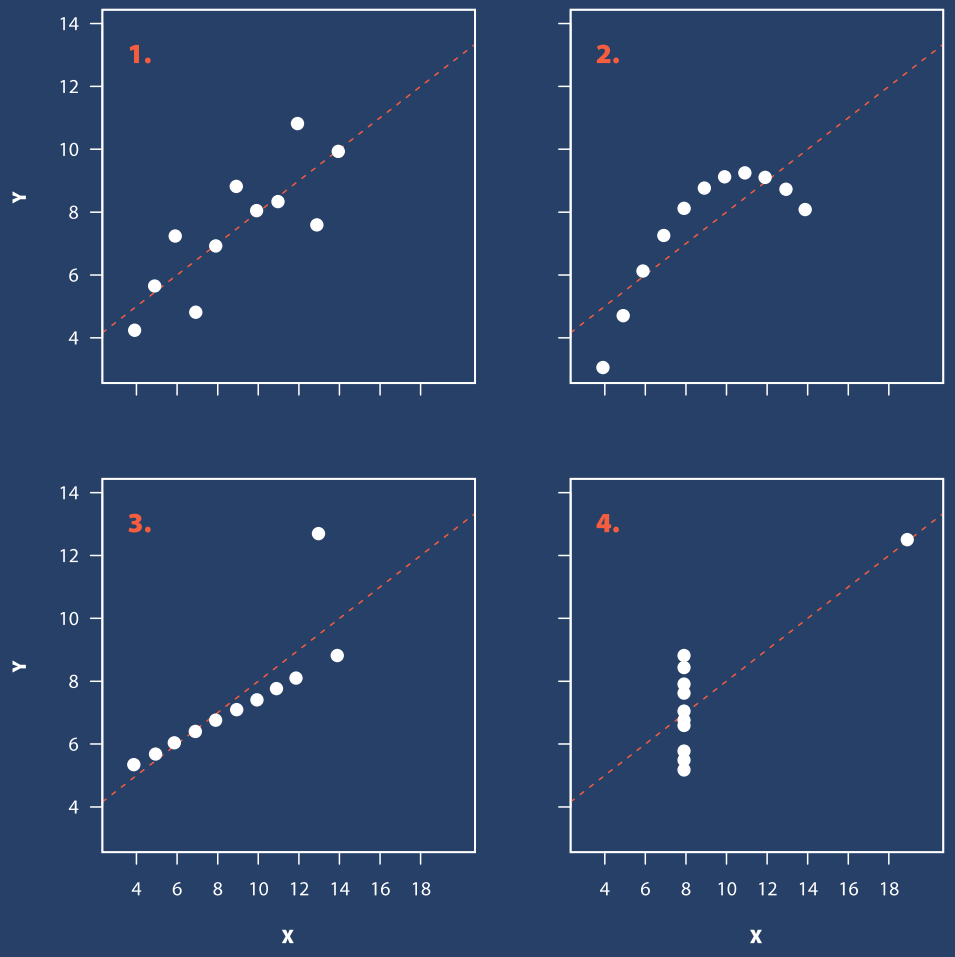

- Papers and proposals
- Talks
- Teach
- Outreach
- Social media + Journalists

**“ To be a successful scientist, you must be an effective communicator”**

# Graphics

Not only data, also patterns

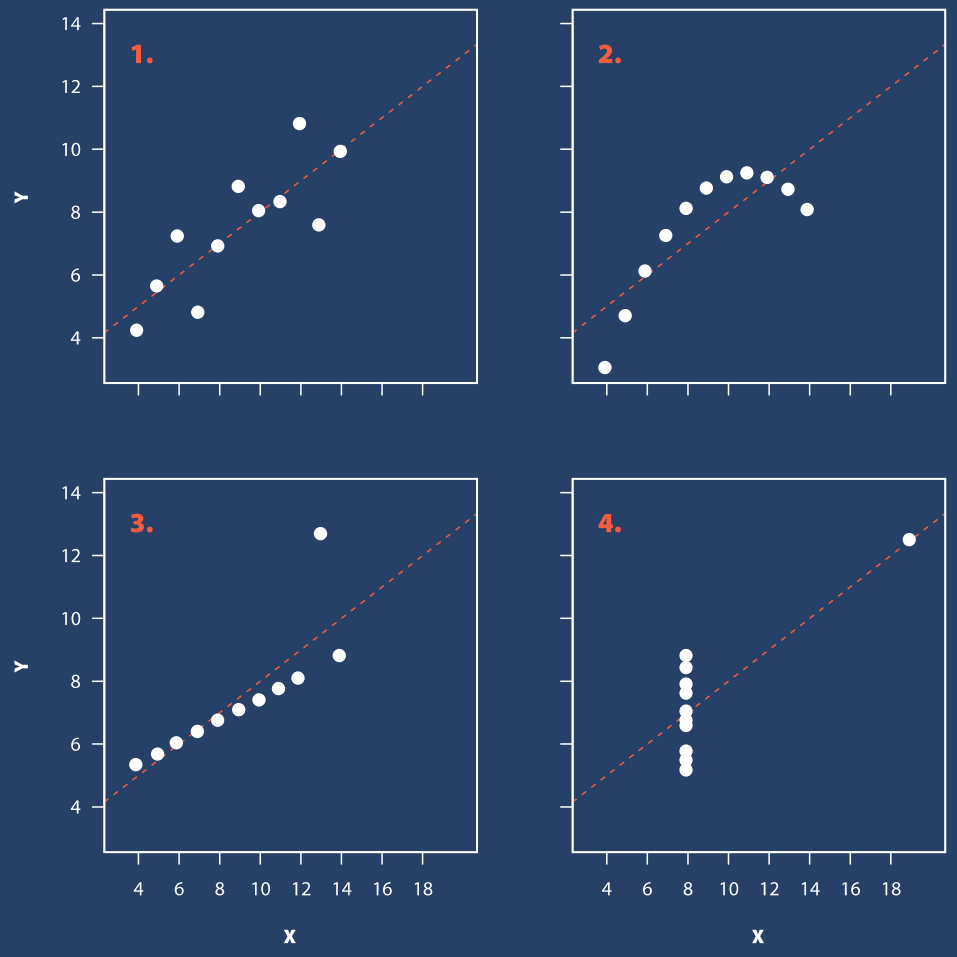
**Anscombe's quartet:**



# Graphics

Not only data, also patterns

**Anscombe's quartet:**




**Same:**

- Number of points
- Mean (X, Y)
- SD (X, Y)
- Regression line
- Standard error of the slope
- Sum of squares
- Residual sum of squares
- Correlation coefficient
- $r^2$

# A good plot should:



Show the data



Tell the truth



Encourage relevant comparisons

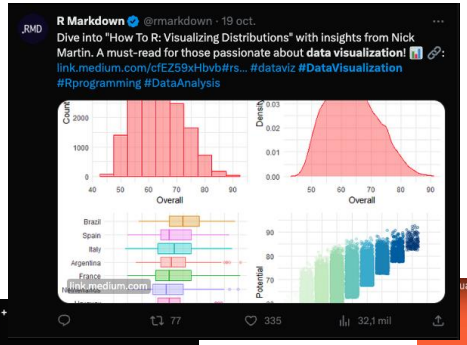


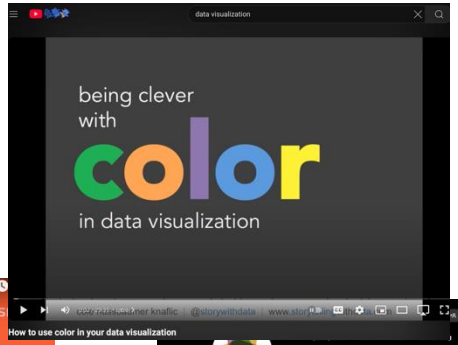
Help the viewer think about the information, not methodology or design

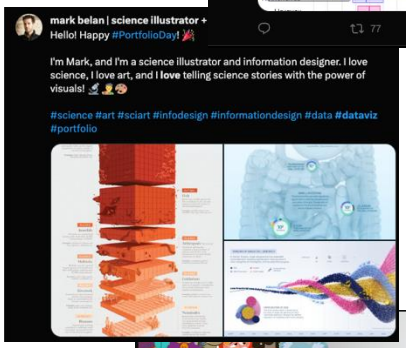


Make large datasets coherent

# DV is popular

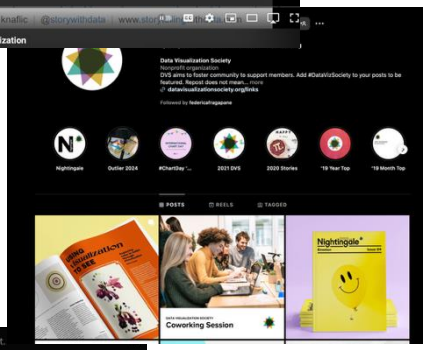


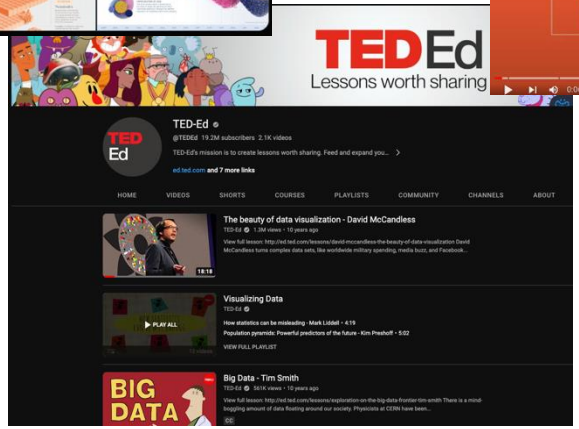



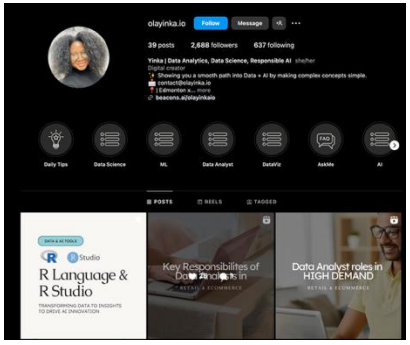



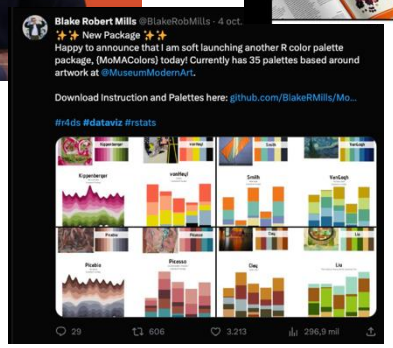






# Guidelines for Data Visualization

Environmental Modelling & Software 26 (2011) 822–827

Contents lists available at ScienceDirect

Environmental Modelling & Software

journal homepage: www.elsevier.com/locate/envsoft




Short communication

## Ten guidelines for effective data visualization in scientific publications

Christa Kelleher*, Thorsten Wagener

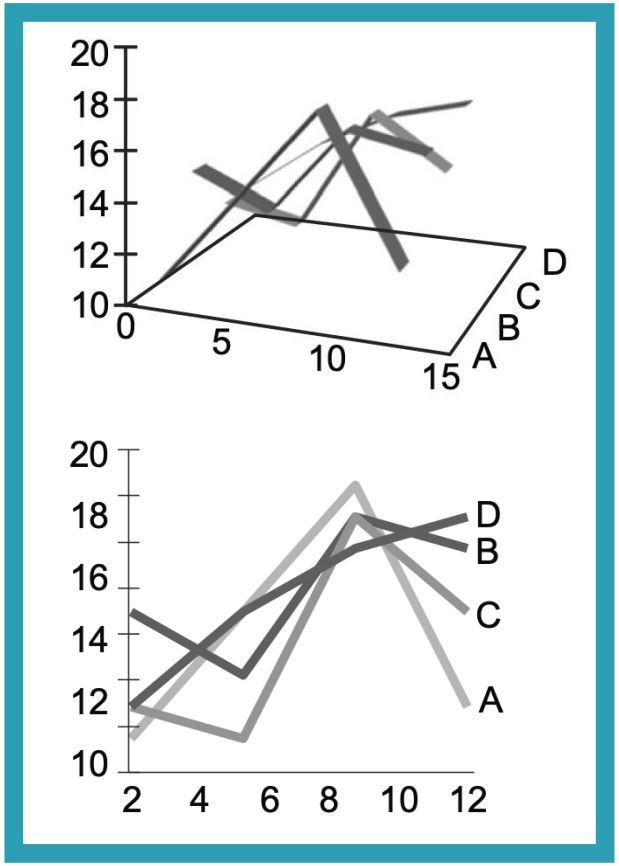
*Department of Civil and Environmental Engineering, The Pennsylvania State University, 212 Sackett Building, University Park, PA 16802, USA*

* Guideline: General principle that can be applied most of the time, but to which there are exceptions

# 1. Less is More

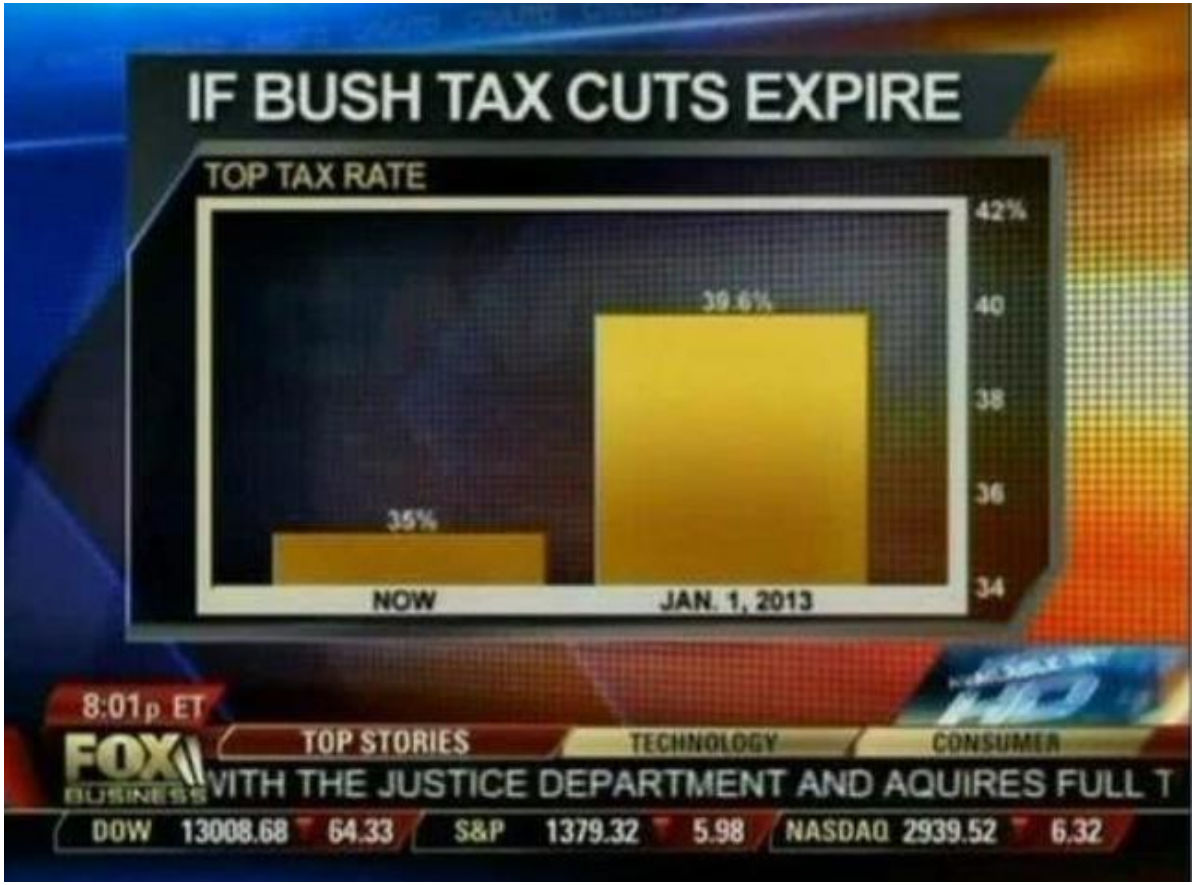
**Create the simplest graph that conveys the information you want to convey**

- Remove redundancy (e.g. using different colors and types of points to show the same information)
- Make sure the reader can distinguish among properties (size, color, line thickness)



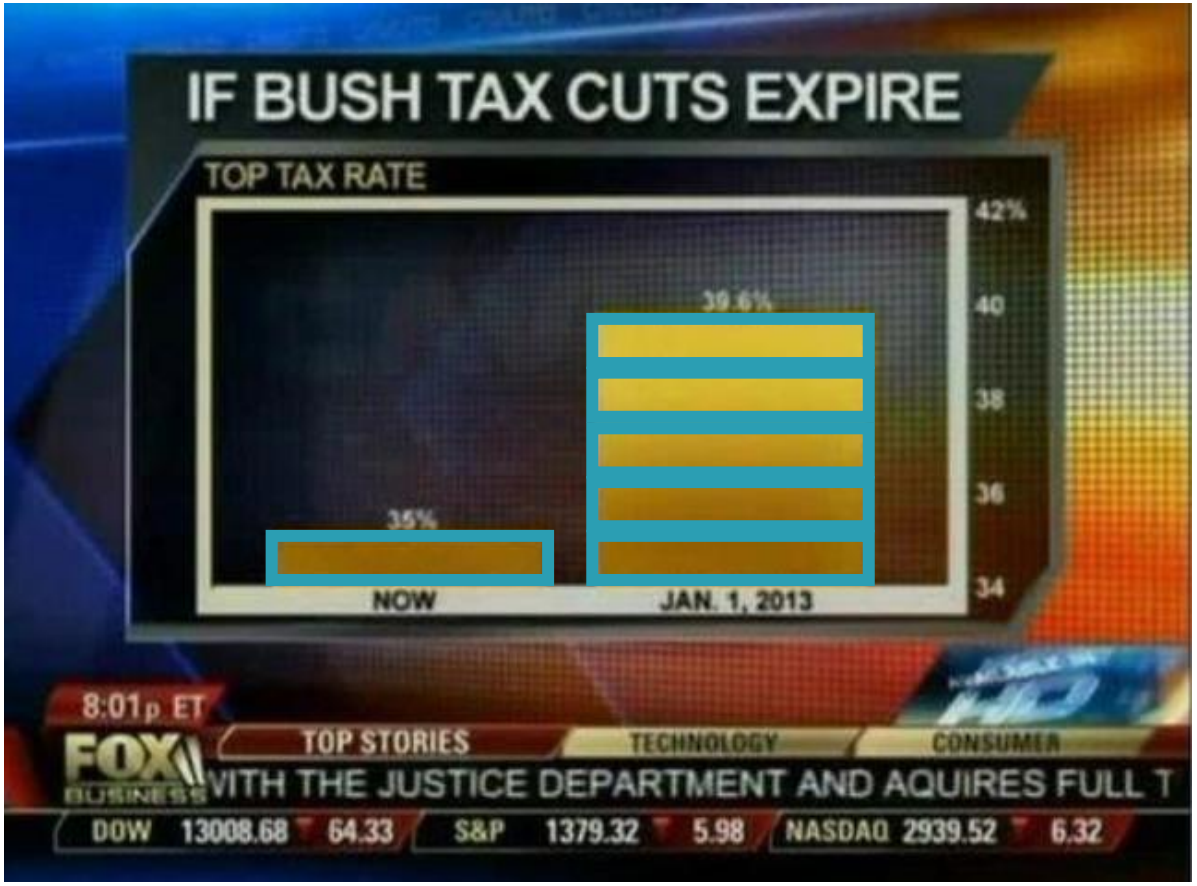
# 2. Be precise

## Select meaningful axis ranges



# 2. Be precise

## Select meaningful axis ranges




After a tax cut expire, there would be a **huge** increase in tax rate

# 2. Be precise

## Select meaningful axis ranges

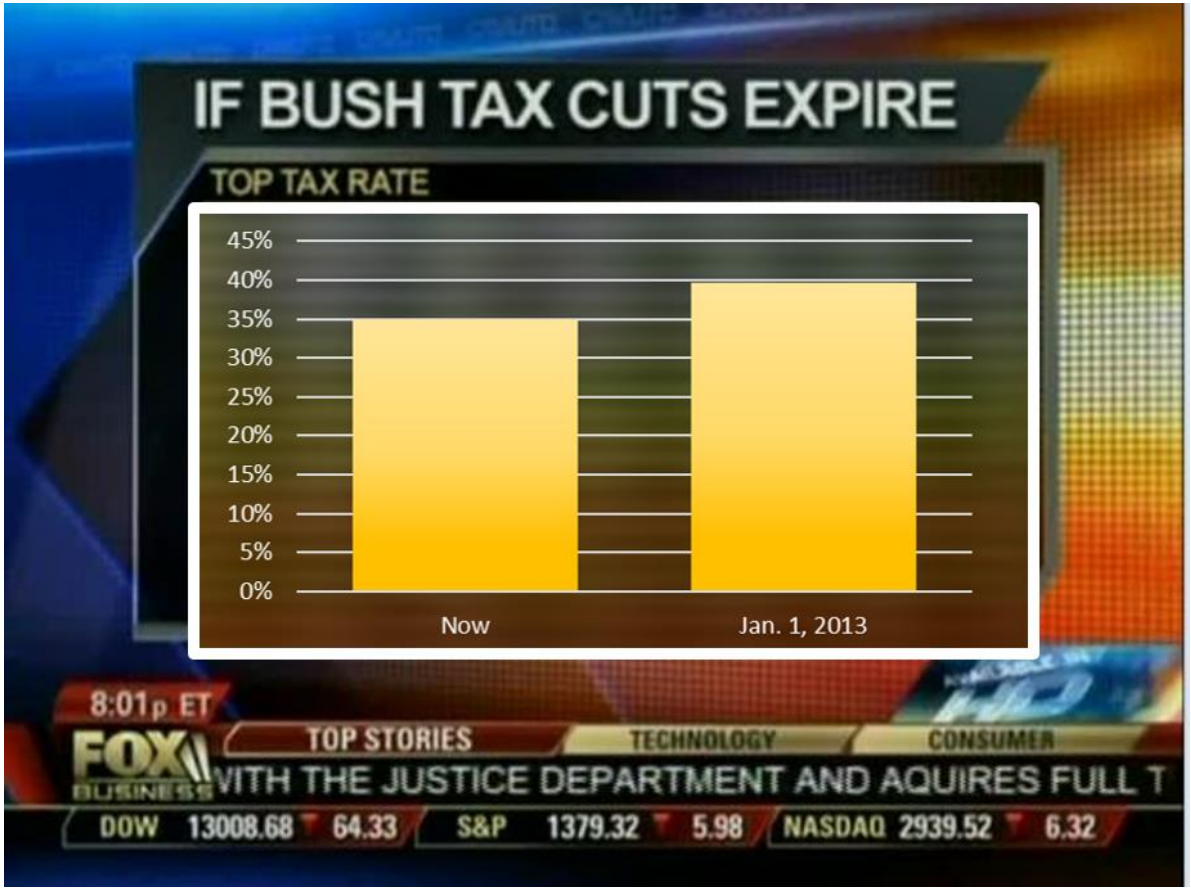



After a tax cut expire, there would be a **huge** increase in tax rate

**BUT**

# 2. Be precise

## Select meaningful axis ranges




After a tax cut expire, there would be a **huge** increase in tax rate

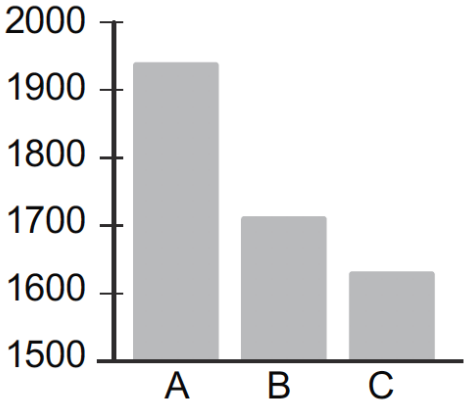
**BUT**

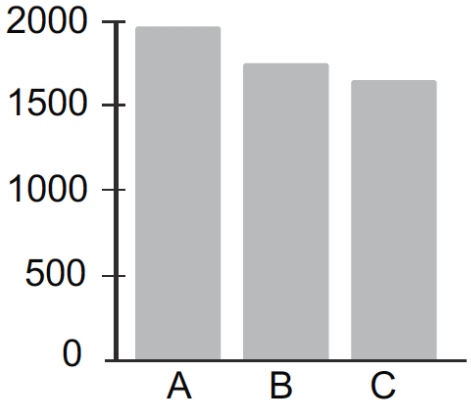
A tax rate of 35% is **not drastically different** from a tax rate of 39.6%

Source: Towards Data Science

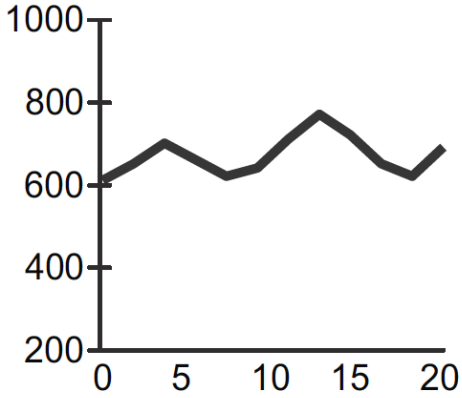
# 2. Be precise

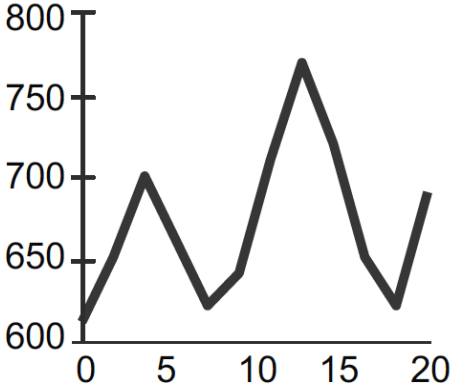
**Selecting the range depends on the graph's purpose and type:**




Absolute magnitudes are important

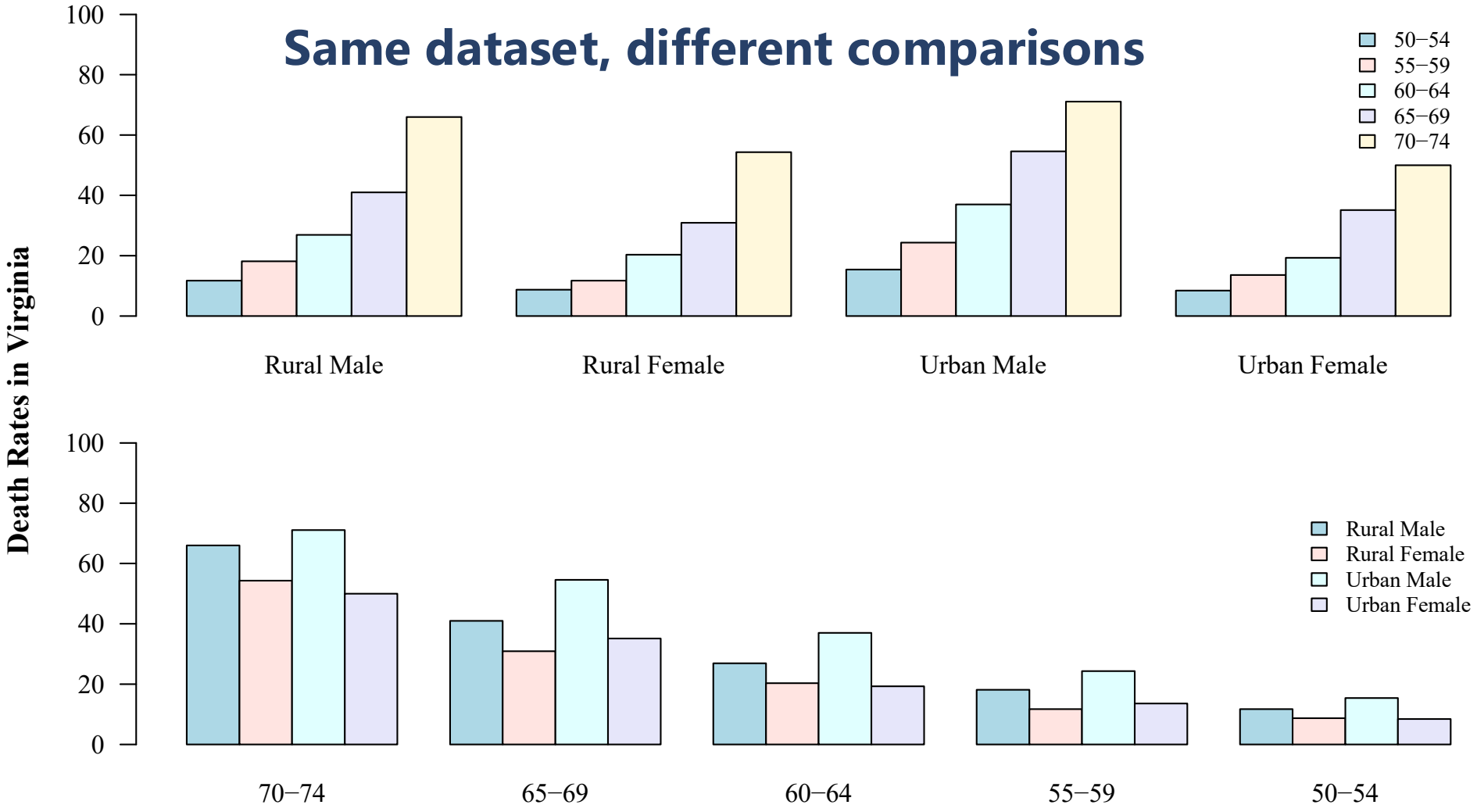


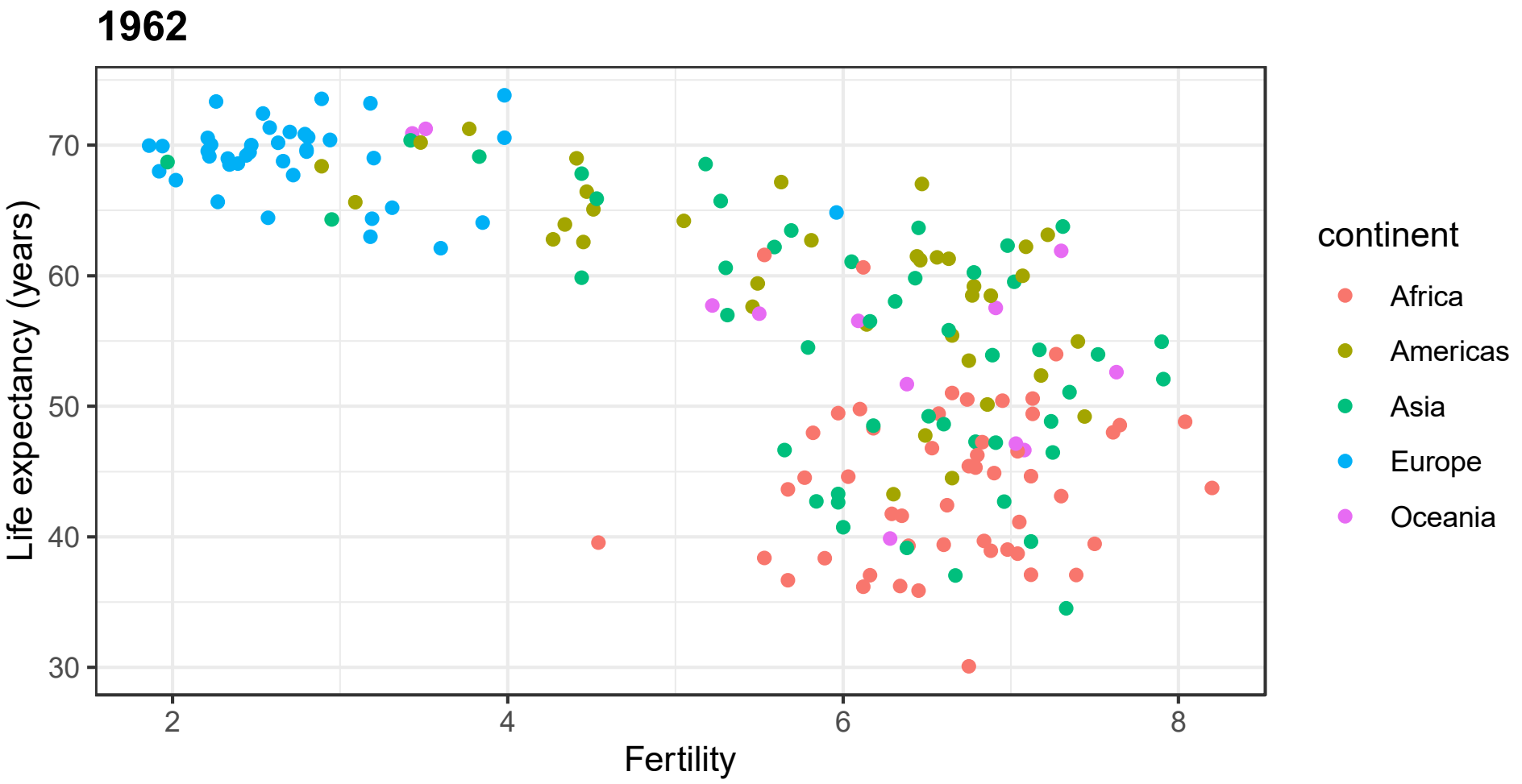

Relative magnitudes are important

Setting the limits of the plot as close as possible to the range of the dataset: improves visualization of variability and avoid white (wasted) space

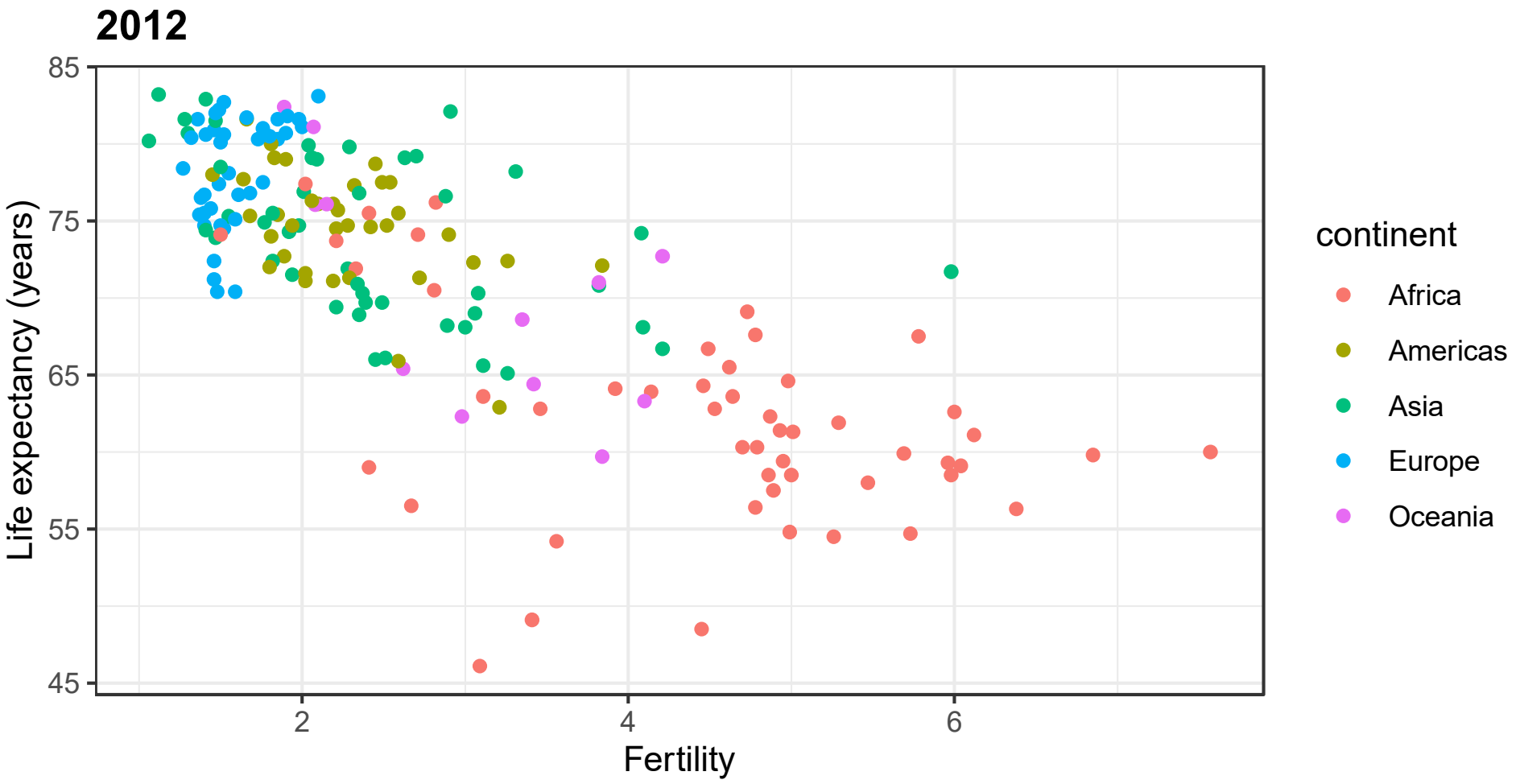
# 3. Relevant comparisons



# 3. Relevant comparisons

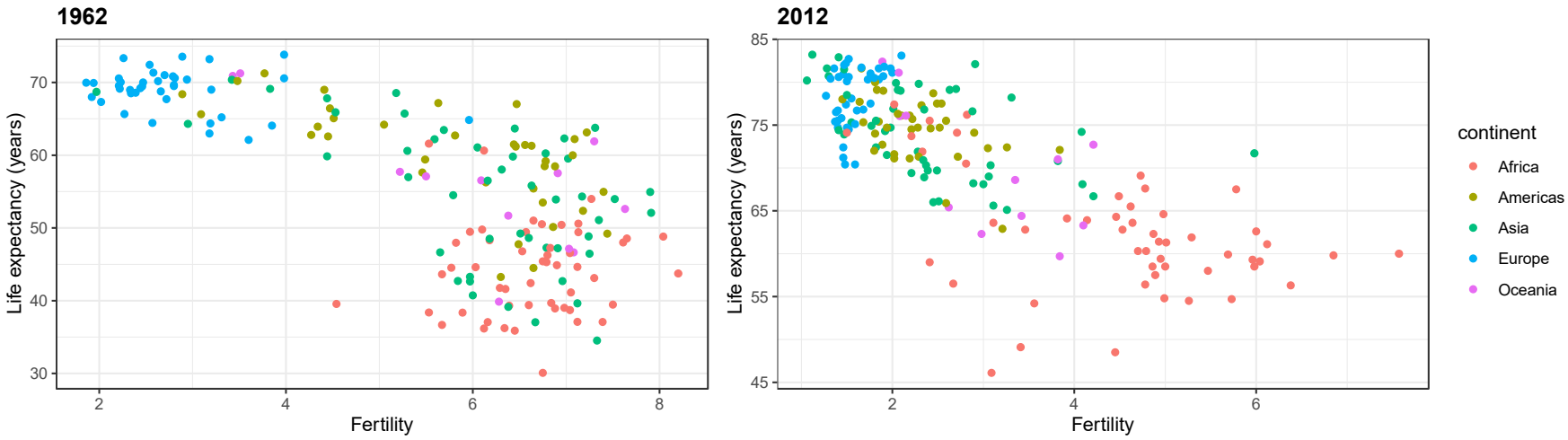


# 3. Relevant comparisons



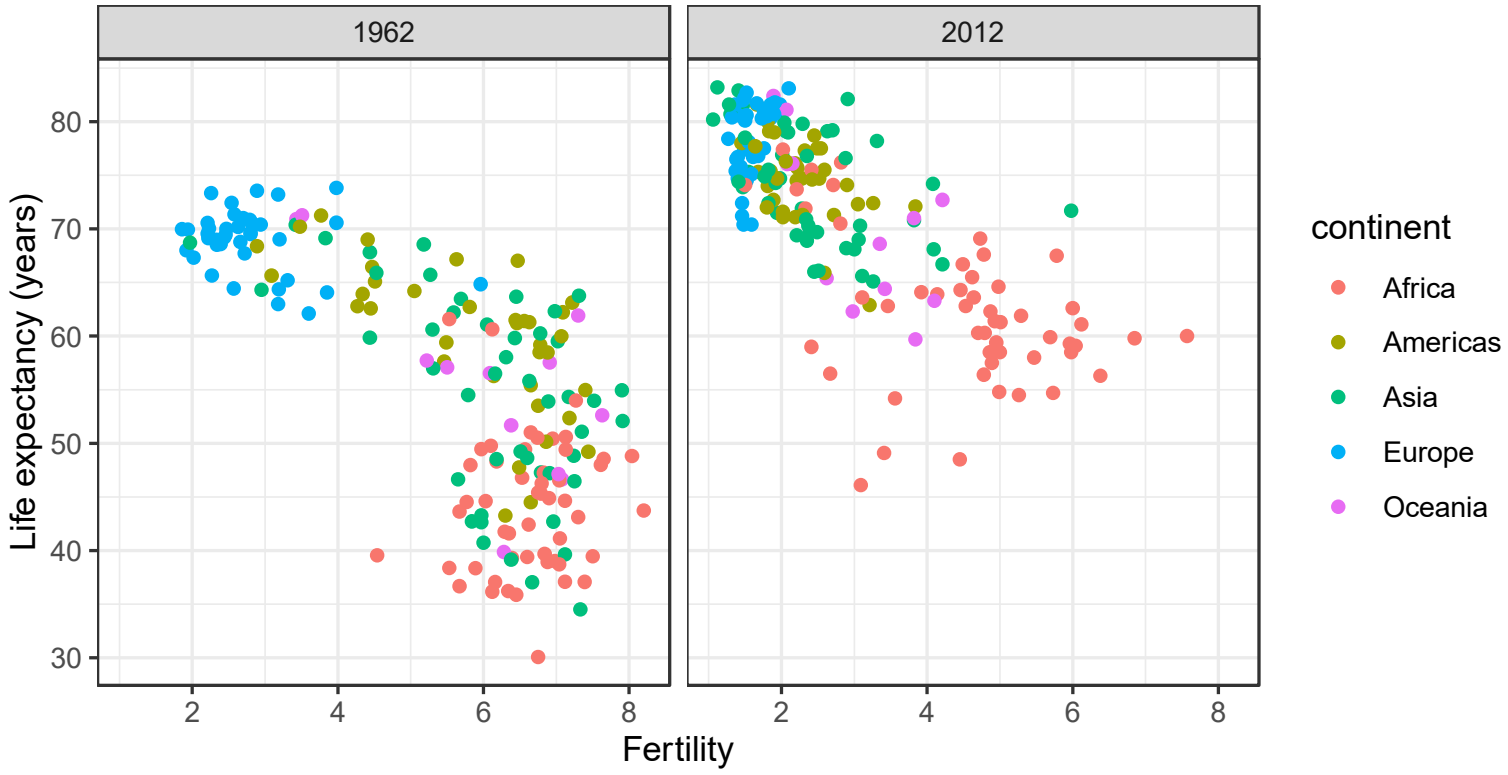
# 3. Relevant comparisons

## It has changed, but not much...



# 3. Relevant comparisons

## Yes, it has changed



# 4. Be careful

## Choosing colors and patterns

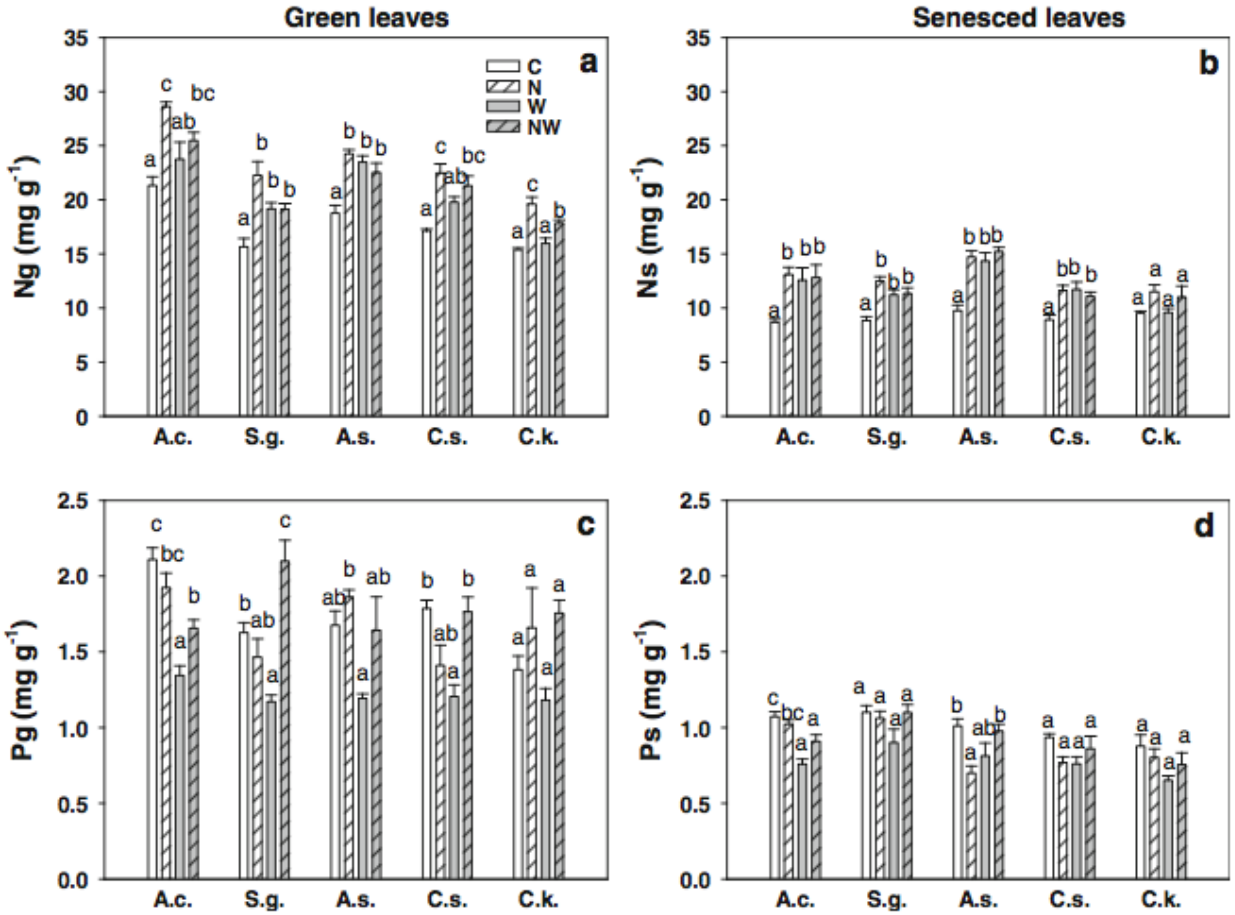


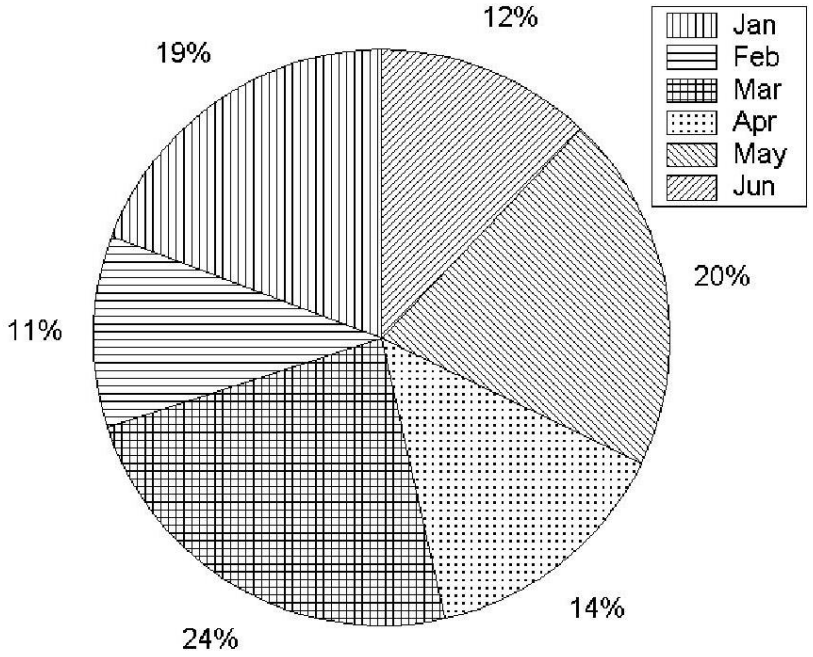

**Fig. 2** Responses of N (**a**, **b**) and P (**c**, **d**) concentrations in green (Ng, Pg) and senesced leaves (Ns, Ps) of five plant species to N and water addition. *Error bars* +1SE. *Bars* within species with *dissimilar letters* denote significant difference ($P < 0.05$). *A.c. Agropyron cristatum*, *S.g. Stipa grandis*, *A.s. Achnatherum sibiricum*, *C.s. Cleistogenes squarrosa*, *C.k. Carex korshinskyi*

*Oecologia, 2012*

# 4. Be careful

## Choosing colors and patterns

**Avoid excessive use of grids, patterns, and decorative forms**



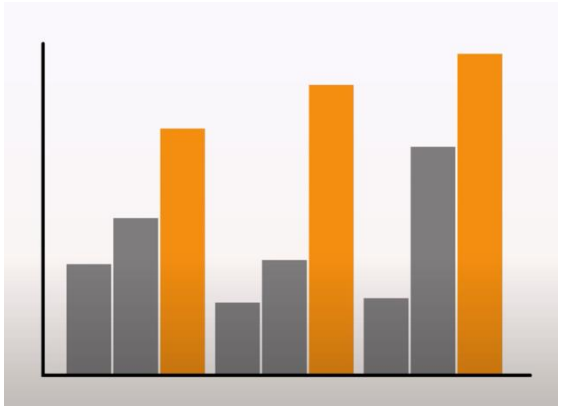
# 4. Be careful

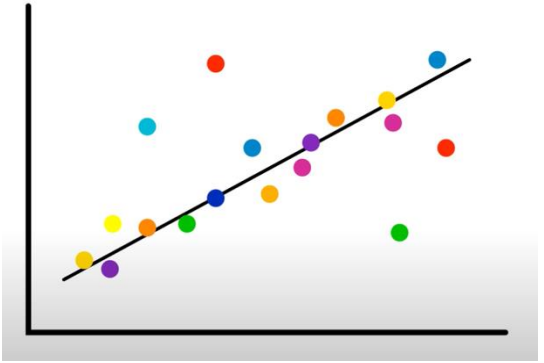
## Choosing colors and patterns

Color can enhance and clarify your figures
(if correctly selected)

or

Color can be confusing, hide the meaning of your data or distort it



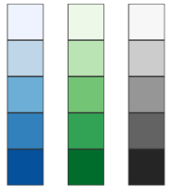


# 4. Be careful

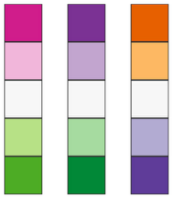
## Choosing colors and patterns

### Use a color scheme that matches the type of data

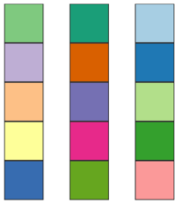
**Sequential:** used for quantitative data varying from low (lighter) to high (darker). Variation of a unique color.



**Diverging:** used to highlight contrasts between low and high values relative to a median value.



**Qualitative:** used mainly for discrete or categorical data. Rapid variation of colors without reference to magnitude.



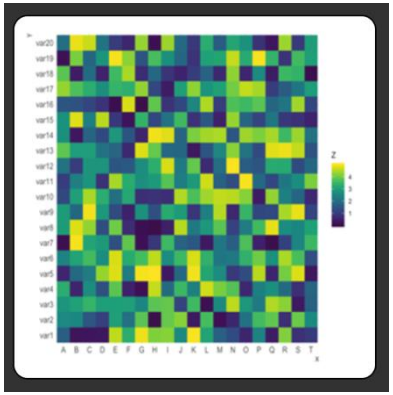
Check ColorBrewer (Website and R package)

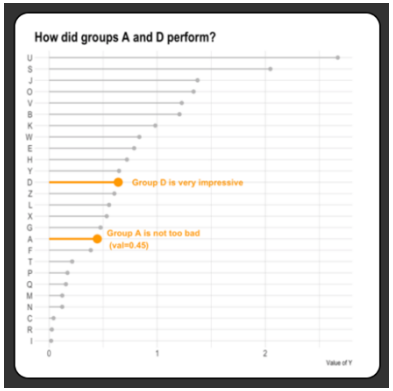
# General suggestions

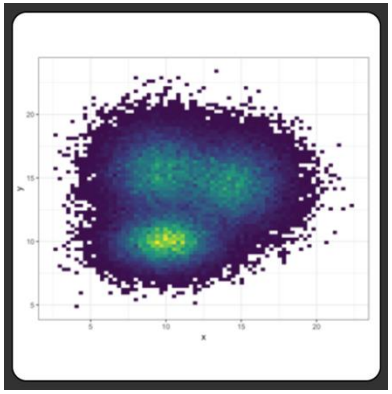
- Use consistent format
- Use readable fonts (size, color, font choice)
- Avoid cluttered legends. Where possible, add labels directly to the elements of the plot
- Make sure the axes are scaled properly
- Axes should be properly labelled, do not forget to specify units
- Plots should be self-explanatory. If needed, include detailed captions that match the graphic
- Choose color wisely (colorblindness = colorblindness)

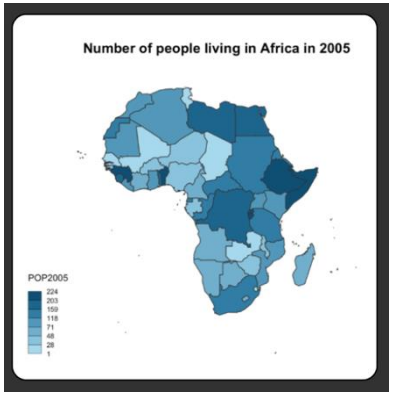
# Plots in R

## Graphic capabilities: one of the strengths in R

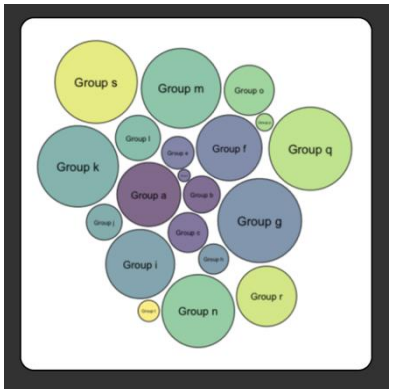


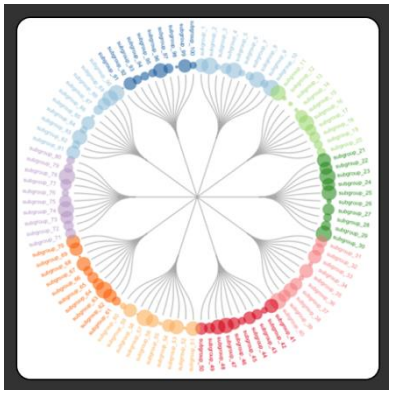


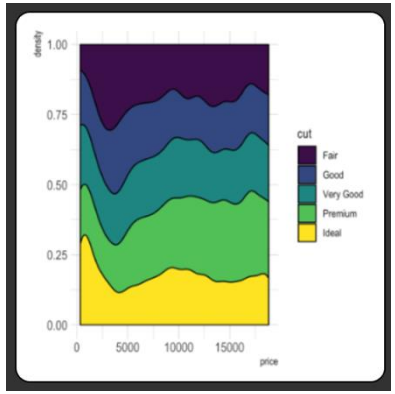









# Plots in R (base)

## Useful functions

**plot( )** # create a new plot of any type
**hist( )** # create a histogram
**lines( )** # add straight lines to an existing plot
**points( )** # add points to an existing plot
**text( )** # add text to an existing plot
**abline( )** # add a line of the form ax + b to an existing plot
**segments( )**, **arrows( )**, **rect( )**, **polygon( )**
# add line segments, arrows, or rectangles to an existing pl
**legend( )** # add a legend to an existing plot

*lines 1 - 42*

# Plots in R (base)

## ? par: Set graphical parameters

[also arguments inside plot ( ), line ( ), etc.]

**pch =** point types
**main =** main title
**col =** set colors
**xlim =** set limits on the x-axis
**ylim =** set limits on the y-axis
**xlab =** set title (label) for the x-axis
**ylab =** set title (label) for the y-axis
**las =** style of axis labels (las = 1, horizontal axes labels)
**lty =** line type (0=blank, 1=solid (default), 2=dashed, 3=dotted, 4=dotdash, 5=longdash, 6=twodash)
**mai =** margin size in inches
**font.lab =** font to be used for x and y labels (1= plain text (default), 2=bold, 3=italic, 4=bold italic)

# Plots in R (base)

## Let's reproduce the Anscombe's quartet plot

**Use the function *pdf ( )*:**

- Starts the graphics device driver for producing PDF graphics (also *svg( )*, *postscript( )*)
- Once you start it, you won't be able to see your plot on RStudio plot window
- Use ***dev.off ( )*** to close the current device. This will save your plot in pdf format, inside your WD

***lines 44 - 131***

# Plots in R (ggplot2)

- "**ggplot2** is a plotting package that makes it simple to create complex plots from data in a data frame"
- "The **ggplot2** package offers a powerful graphics language for creating elegant and complex plots. Its popularity in the R community has exploded in recent years"
- "Mastering the **ggplot2** language can be challenging"
- "**ggplot2** is the most elegant and aesthetically pleasing graphics framework available in R"

# Plots in R (ggplot2)

**It is very different from base graphics**

**Basic Structure:**

1. The setup
2. The layers
3. The labels
4. The theme
5. The facets

# Plots in R (ggplot2)

1. The setup: tell ggplot what dataset to use and set x and y variables

   Use the functions

   **ggplot ( ):** Create a new ggplot

   and

   **aes( ):** Set the aesthetics mappings

   *"How variables are mapped to the visual properties of the geoms"*

```
p <- ggplot(iris, aes(x=Sepal.Width, y=Sepal.Length))
```

# Plots in R (ggplot2)

2. Add the layers (also "geoms" or geometric objects)
   using the + symbol

```
+ geom_point( )
```

There are many **geom_** functions: geom_boxplot, geom_abline, geom_histogram, geom_errorbar, etc.
Check the *ggplot* Reference page

***lines 149 - 151***

# Plots in R (ggplot2)

3. **Add some labels:** you might want to add the plot's main title and perhaps change the X and Y axes titles.

   This can be accomplished using the labs ( ) layer

```
+ labs(title="Scatterplot",
       x = "Sepal width (cm)",
       y = "Sepal length (cm)")
```

*lines 154 - 159*

# Plots in R (ggplot2)

4. **Change the *theme*:** Themes control the display of all non-data elements of the plot (font color, font size, background, gridlines, etc.)

```
+ theme ( )
```

For more: click here

*lines 161 - 169*

# Plots in R (ggplot2)

5. **The facets:** add panels to display subsets of the data

```
+ facet_wrap(~Species, ncol=3)
```

*lines 171 - 178*

# Plots in R (ggplot2)

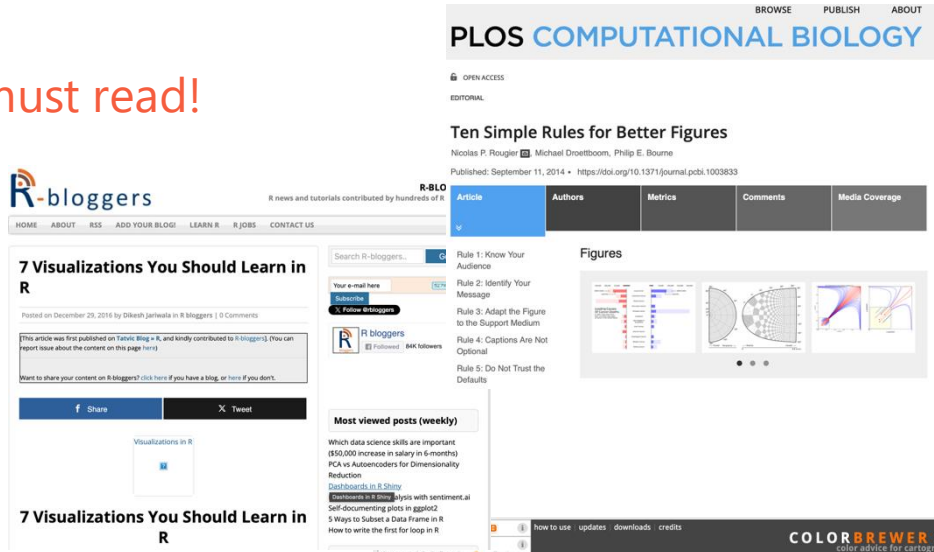
6. **And finally, save the plot**

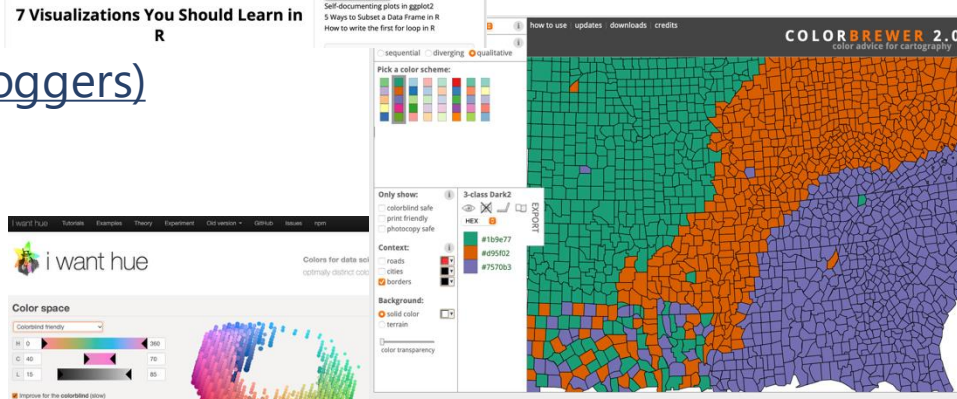
ggsave( “Myplot.pdf”)

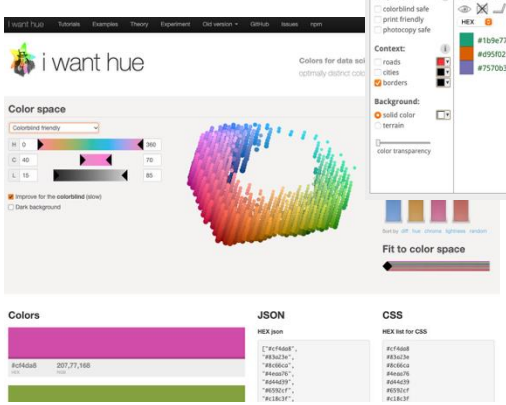
*lines 180 - 181*

# Useful Online Resources:

- Ten simple rules for better figures - must read!
- From Data-to-viz
- R Graph Gallery
- 7 visualizations you should learn (r-bloggers)
- Selected tools for Data visualization
- Viridis color maps
- R ColorBrewer

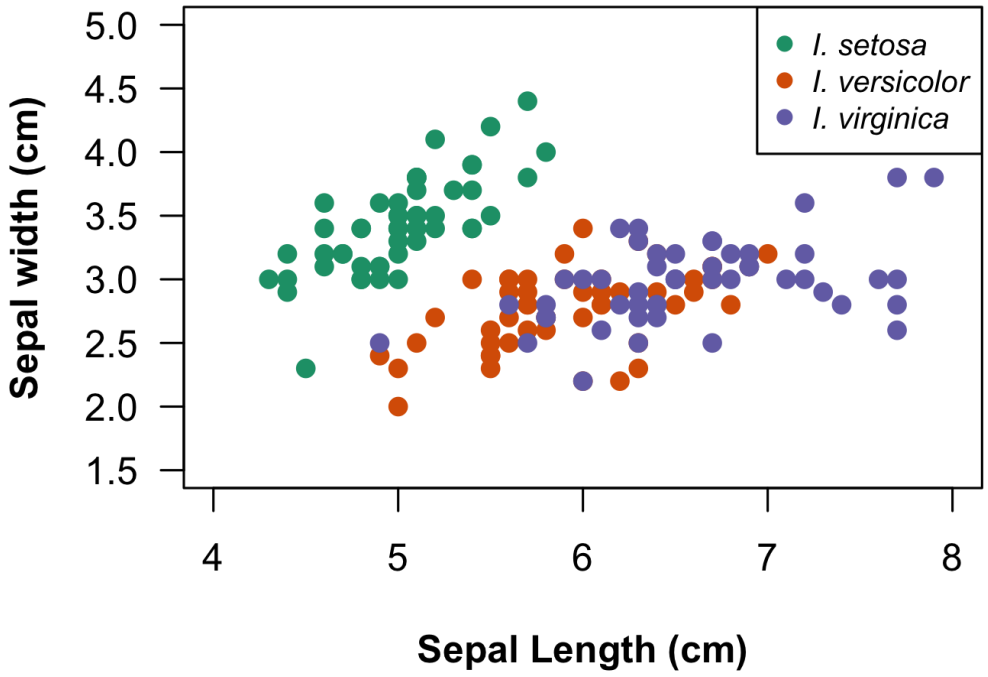


# Let's make a plot

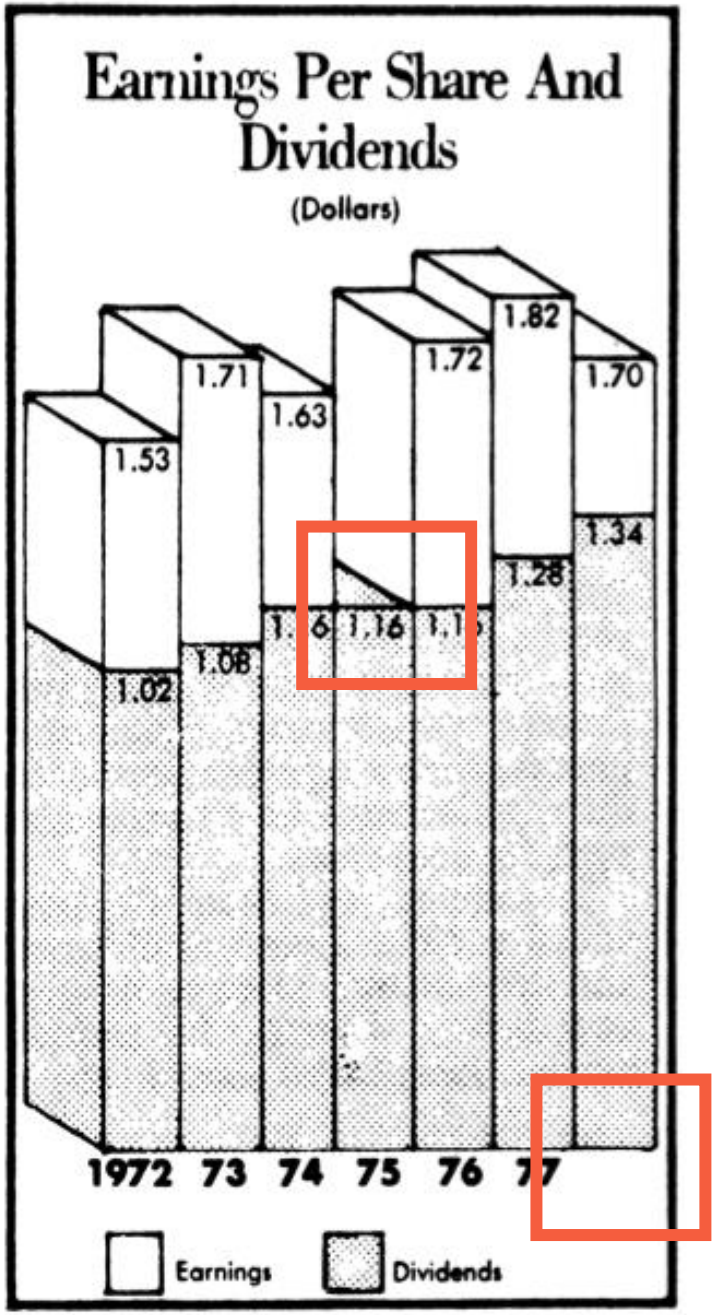
We will work on the iris dataset (again!) and make a scatterplot of sepal width (y-axis) vs. sepal length (x-axis). We will look at differences among species using points with different colors. Your plot should look like this:




Help: Explore the R Graph Gallery and R ColorBrewer

Challenge: Try to make the same plot in *Base R* and in *ggplot*

# Additional Examples




# Less is more

Increasing the number of dimensions to a graph makes it more confusing

**Even for the person who made it!**

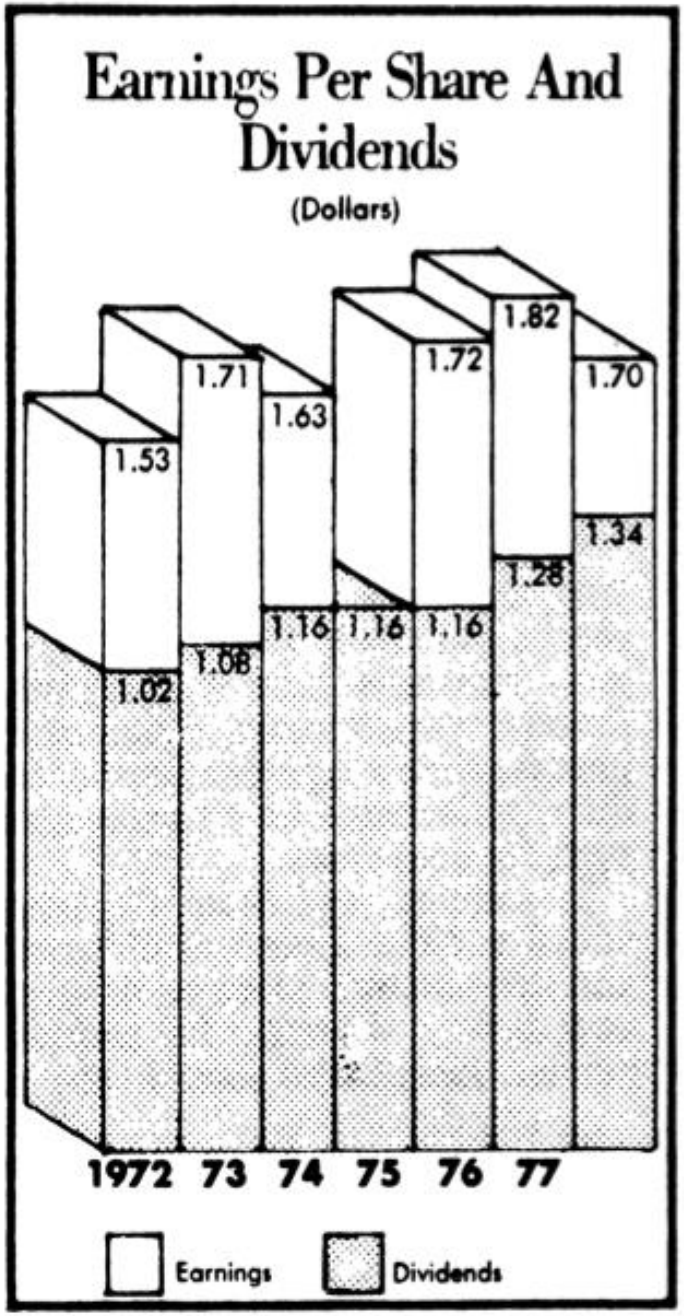



*Figure 23.*
*(© 1979, The Washington Post).*

# Less is more

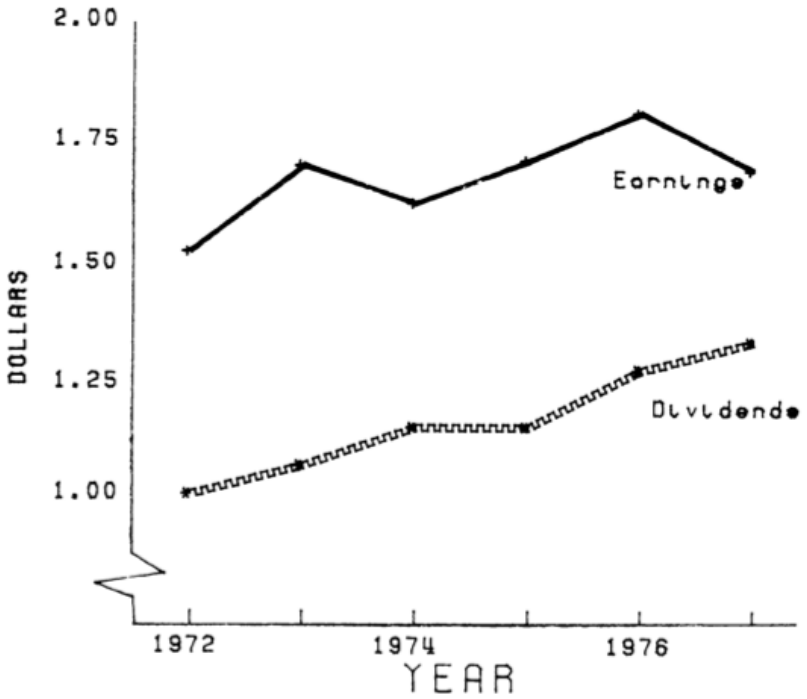
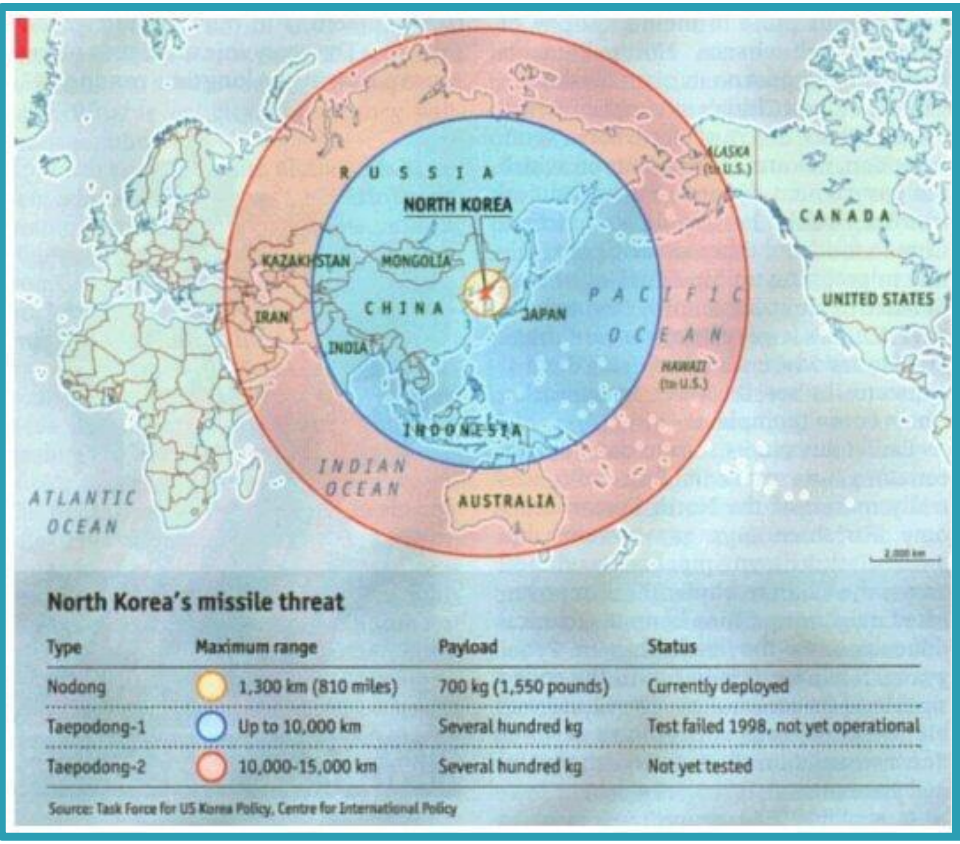
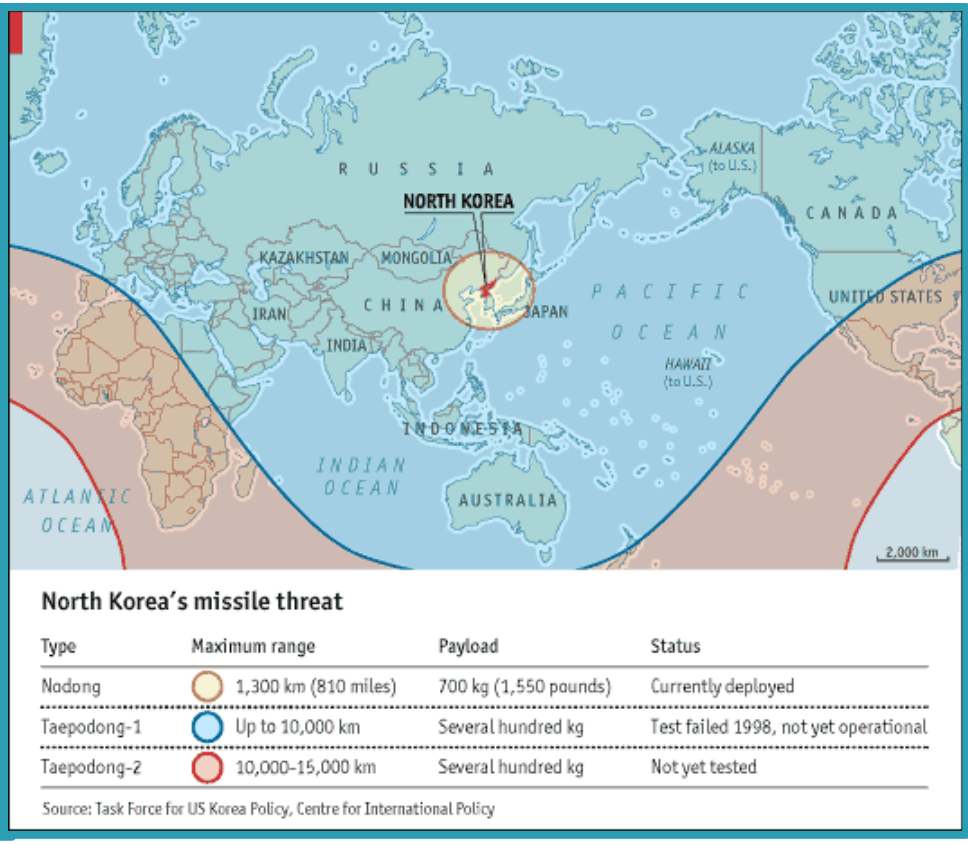



*Figure 24. Data from Figure 23 redrawn simply*

# Precision and accuracy




*The Economist*

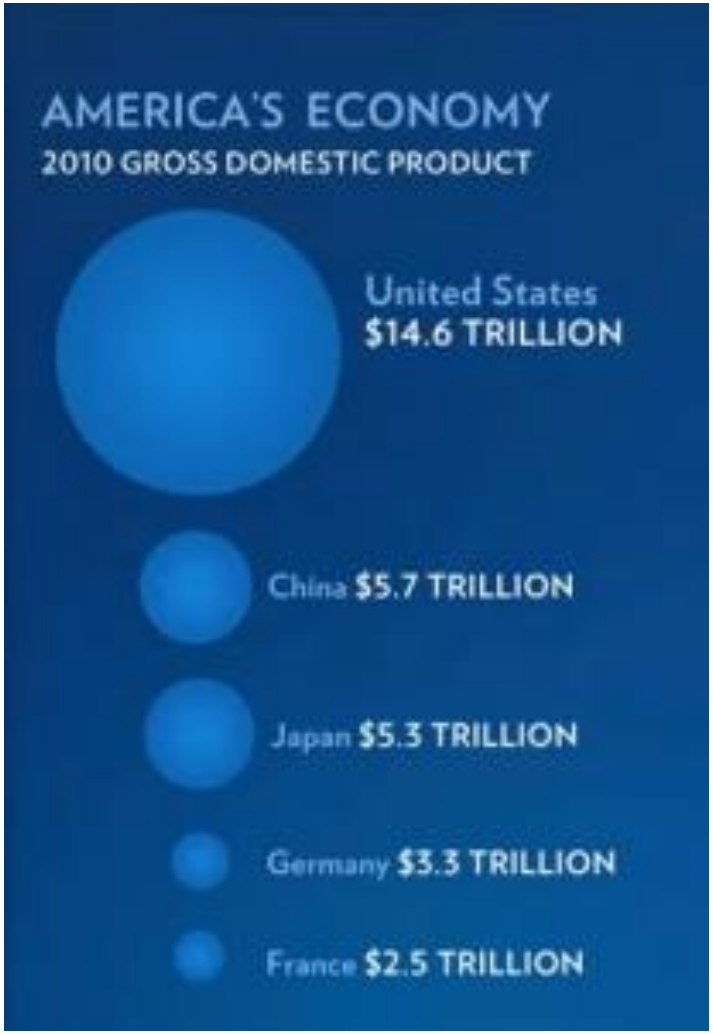



*The Economist, a week later*

# Precision and accuracy

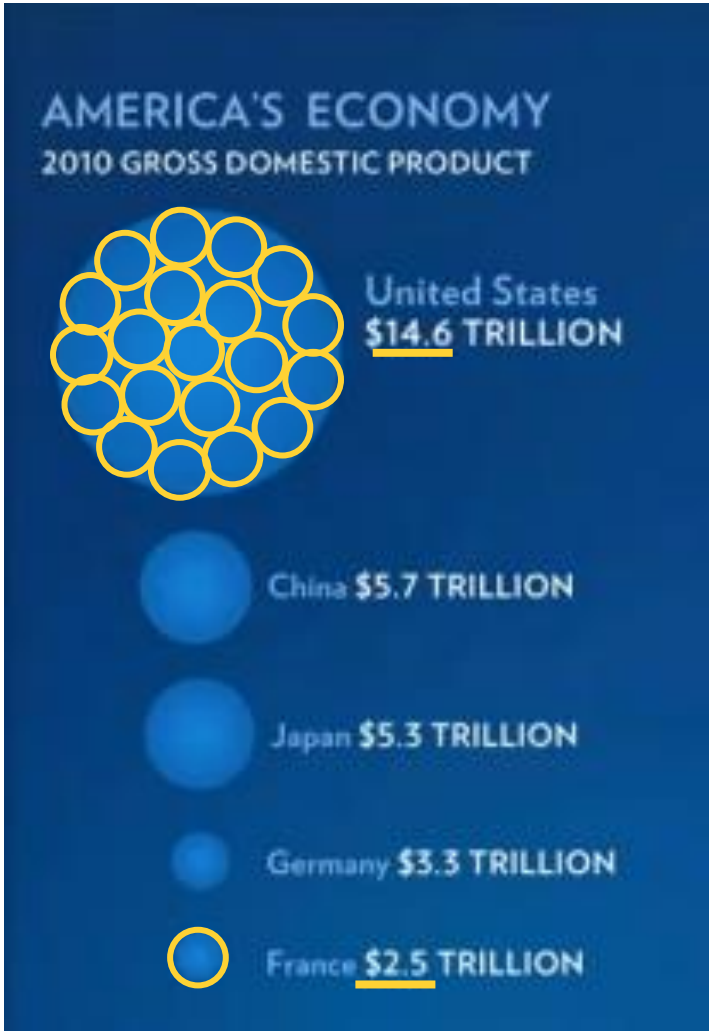
- Do not take data out of context
- Show variation in the data
- Label as necessary to make things clear
- Use appropriate (and comparable) units of measure

# Precision and accuracy



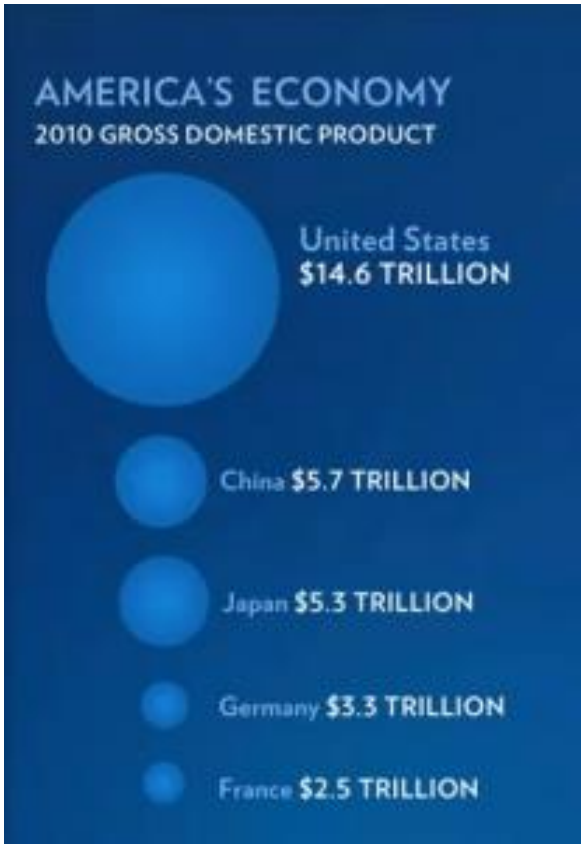

This plot may deceive viewers because the differences in areas don't correspond to the differences in the numbers presented

# Precision and accuracy




This plot may deceive viewers because the differences in areas don't correspond to the differences in the numbers presented

# Precision and accuracy





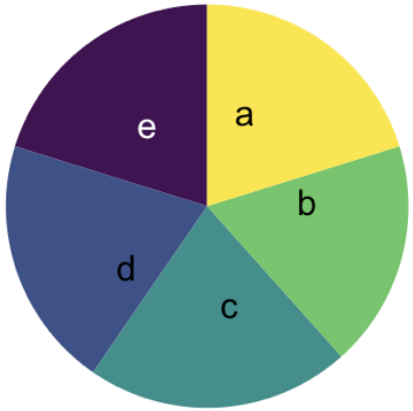


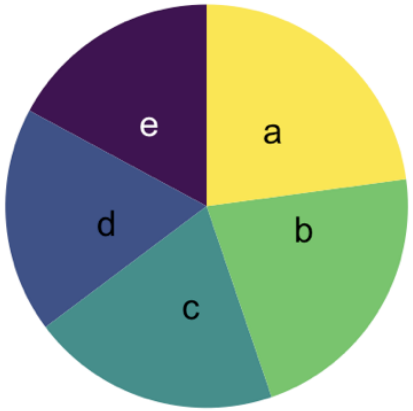
**Corrected**

# Precision and accuracy

## Avoid pie charts




**Humans are bad at reading angles. On each pie, which group is the biggest one? Try to order them by value.**

# Precision and accuracy

